CorticalFields: a genuinely novel library at the intersection of five underexploited gaps

GP-based surprise maps on cortical surfaces for epilepsy are feasible, deeply novel, and clinically needed. No existing tool computes vertex-wise probabilistic anomaly scores on the cortical manifold using geodesic-aware kernels — every current normative model treats each vertex or ROI independently, discarding the spatial structure of the cortex. The mathematical foundations are mature (Borovitskiy et al. 2020; GeometricKernels library), the clinical motivation is strong (ENIGMA-Epilepsy shows widespread cortical thinning in MTLE-HS that no tool quantifies at the individual level), PubMed Oxford Academic and at least five of CorticalFields' proposed modules occupy territory that is either completely unexplored or only partially bridged between communities. This report maps the state-of-the-art across all ten research areas, identifies what has and has not been done, and positions CorticalFields precisely within the gap structure.

Matérn GPs on manifolds are theoretically solved but clinically unapplied

The foundational theory for CorticalFields already exists. **Borovitskiy, Terenin, Mostowsky, and Deisenroth (NeurIPS 2020)** proved that Matérn kernels on compact Riemannian manifolds can be constructed via spectral decomposition of the Laplace-Beltrami (LB) operator: arXiv $k(x, y) = \sum_\ell \Phi_{\nu,\kappa}(\lambda_\ell)\, f_\ell(x) f_\ell(y)$, where f_ℓ are LB eigenfunctions and Φ is a spectral density controlled by smoothness ν and lengthscale κ. Critically, Feragen et al. (CVPR 2015) showed that naïvely replacing Euclidean distance with geodesic distance in the standard Matérn formula does **not** yield a valid positive-definite kernel on curved manifolds ACM Digital Library — the spectral approach is the only correct path.

This theory has been implemented in the **GeometricKernels** library (Mostowsky et al., JMLR 2025), which supports heat and Matérn kernels on manifolds, meshes, and graphs with backends for PyTorch, JAX, and TensorFlow, PyPI and provides a GPyTorch frontend. Snyk On meshes, the library uses the robust Laplacian (Sharp & Crane, SGP 2020 best paper) and computes kernels via truncated eigendecomposition. Extensions to graphs (Borovitskiy et al., AISTATS 2021), vector fields (Hutchinson et al., NeurIPS 2021), and unknown manifolds (Fichera et al., NeurIPS 2023) have followed. arXiv Sml-group

The critical gap is application. The closest work to CorticalFields is a 2025 paper in *Imaging Neuroscience* using Riemannian diffusion kernels on native cortical meshes — but only for kernel density estimation of tractography endpoints, (MIT Press) not GP regression or anomaly detection. (bioRxiv) Consagra et al. (Biometrics 2024) fitted GP priors on the cortical surface for group-level fMRI regression using Vecchia approximation (PubMed Central) (arXiv) on **~30k vertices** with great-circle distance on the sphere, (arXiv) (PubMed Central) and Mejia et al. (JASA 2020) used INLA with SPDE-approximated Matérn fields for cortical fMRI. (arXiv) He et al. (arXiv 2024) achieved O(n) scaling for heat kernel GPs on manifolds using truncated SVD of the graph Laplacian transition matrix, demonstrating scalability on HCP fMRI data. None of these papers perform anomaly detection, normative modeling, or work with structural MRI morphometric features on cortical surfaces.

Scalability at 150k vertices is achievable. Exact GP inference ($O(n^3)$) is impossible at this scale, but the spectral Matérn approach reduces the problem: computing L = 200–1000 eigenpairs of the sparse mesh Laplacian is feasible in minutes with ARPACK, after which kernel evaluation costs O(L) per vertex pair. Combining this with SVGP (m = 500–1000 inducing points, costing $O(nm^2)$ per step) (NTU Singapore) or reduced-rank approximations (He et al. 2024, O(n)) makes the pipeline tractable. KeOps provides GPU-accelerated kernel matrix-vector products with O(n) memory (Kernel-operations) for up to **1 million points**, though for spectral Matérn kernels the bottleneck is eigenpair computation rather than kernel MVMs.

Every normative model ignores cortical geometry — and this matters

The normative modeling ecosystem, dominated by **PCNtoolkit** (Marquand, Wolfers, Rutherford et al.), operates entirely in a mass-univariate fashion: a separate, independent model is fit for each brain region or vertex. The toolkit's primary method — warped Bayesian linear regression with SinhArcsinh likelihoods (Fraza et al., NeuroImage 2021) — handles non-Gaussianity and site effects elegantly, (ScienceDirect) but **no spatial information is used**. Pre-trained models covering ~58,000 subjects exist for cortical thickness and surface area, (PubMed Central) but only at the parcellated (Desikan-Killiany/Destrieux) level with **68–150 ROIs**.

The most direct competitor is **Spectral Normative Modeling** (SNM, 2025), which uses LB eigenmodes for dimensionality reduction, achieving vertex-wise normative z-scores with a **98.3% speedup** over fitting independent vertex models. SNM implicitly encodes spatial structure through the eigenmode basis and was validated on Alzheimer's detection. However, it assumes Gaussianity, does not produce a full posterior predictive distribution, and does not compute surprise (negative log-predictive density). It uses eigenmodes for compression, not as a GP kernel.

Zabihi et al. (MICCAI 2018) proposed scalable multi-task GP regression for normative modeling of fMRI with Kronecker-factored spatial covariance — the only published attempt to model spatial correlation in normative modeling. But it was applied to volumetric fMRI with data-driven ROI correlations, never to cortical surfaces or geodesic distances.

Three things are simultaneously novel about CorticalFields' normative approach:

- **No normative model uses a geodesic/Matérn kernel on the cortical surface.** The mathematical machinery exists (Borovitskiy 2020, GeometricKernels) but has never been connected to normative modeling. PubMed Central MIT Press
- **No method computes pointwise surprise (−log p(y|x, model)) on cortical surfaces** from a probabilistic model conditioning on spatial neighbors. Current methods produce z-scores (assuming Gaussianity, ignoring spatial correlation) or deep learning reconstruction errors (not calibrated probabilistic scores).
- **No vertex-wise normative model has been applied to epilepsy.** All normative modeling in epilepsy operates at ROI level. Oxford Academic Brain MoNoCle (Little et al., Imaging Neuroscience 2025) validated ROI-level normative models on mTLE, MIT Press and Mito et al. (Brain 2025) wrote a comprehensive review explicitly calling for normative modeling in epilepsy Oxford Academic — but vertex-wise implementation remains absent.

Spectral shape descriptors from computer graphics have barely crossed into clinical neuroimaging

A striking gap exists between the geometry processing community and clinical brain imaging. **Heat Kernel Signatures** (Sun et al. 2009), the canonical per-vertex shape descriptor from computer graphics, have essentially **never been applied to adult cortical surfaces** in clinical neuroimaging. Only a single recent study on fetal brain development (circa 2024–2025) computed HKS and WKS on cortical surfaces, with the authors explicitly noting these are "rarely applied in the medical field." ResearchGate **Wave Kernel Signatures** (Aubry et al. 2011) are virtually absent; the only neuroimaging-adjacent application is manifold alignment for MR image reconstruction PubMed (Balfour et al., IEEE TMI 2019).

What has crossed the boundary are **global** spectral descriptors. **ShapeDNA** (Reuter et al. 2006) uses ordered LB eigenvalues as isometry-invariant shape fingerprints, and **BrainPrint** (Wachinger et al., NeuroImage 2015) extends this to ensembles of brain structures, ScienceDirect achieving >99.8% subject identification accuracy. In epilepsy specifically, Shishegar et al. (Neurological Sciences 2022) used LB eigenvalues and eigenfunctions for hippocampal shape classification, achieving **93% diagnostic accuracy** and **86% lateralization** in temporal lobe epilepsy — outperforming volumetric analysis. ResearchGate Harms et al. (Epilepsy & Behavior 2022) applied BrainPrint to hippocampal shape asymmetry in TLE.

However, all epilepsy spectral work focuses on subcortical (hippocampal) shape. **No spectral shape descriptor of any kind has been applied to cortical surface analysis in epilepsy.** The gap between global descriptors (eigenvalues only) and vertex-wise descriptors (HKS, WKS, GPS) for individual anomaly detection remains completely unfilled. Kim et al. (NeuroImage 2014) used spectral graph wavelets on cortical thickness for Alzheimer's group analysis, PubMed Central demonstrating feasibility but not individual-level anomaly detection.

CorticalFields' inclusion of HKS/WKS as vertex-wise cortical features for anomaly detection would be the first bridge between these communities — leveraging a decade of geometry processing theory that neuroimaging has overlooked.

Geodesic computation and the LB eigenpair pipeline are production-ready

The computational infrastructure for CorticalFields' core pipeline is mature. For geodesic distance computation, **potpourri3d** wraps geometry-central's heat method (Crane et al. 2013/2017) in a clean Python API, GitHub PyPI while **gdist** (used in The Virtual Brain) provides fast marching on meshes PyPI GitHub and is integrated into PyTorch Geometric's `geodesic_distance()` utility. PyTorch Geometric **LaPy** (Deep-MI group) is the most brain-focused tool: a pure-Python FEM solver for the LB operator with ShapeDNA computation, heat equation geodesics, and optional CHOLMOD acceleration via scikit-sparse. The **robust-laplacian** library (Sharp & Crane, SGP 2020) builds numerically stable Laplace matrices even on nonmanifold or poor-quality meshes via intrinsic Delaunay triangulation. GitHub GitHub

For CorticalFields, full pairwise geodesic distance matrices at 150k vertices are unnecessary and infeasible ($O(N^2)$ storage ≈ **180 GB** for float64). The spectral Matérn approach elegantly sidesteps this: one computes the top L eigenpairs of the sparse N×N mesh Laplacian (feasible for $L \leqslant 1000$ with iterative eigensolvers), then evaluates kernels entirely in the spectral domain. The 2025 *Imaging Neuroscience* paper recommends ~1000 LB eigenpairs as default for Riemannian diffusion kernels on cortical surfaces, confirming practical feasibility. MIT Press PubMed Central Nyström approximation with landmark-based geodesics provides an alternative for methods that do require pairwise distances. Andy Jones

The recommended pipeline is: FreeSurfer surface → **robust-laplacian** or **LaPy** for mesh Laplacian → ARPACK eigensolver for L eigenpairs → **GeometricKernels** for spectral Matérn kernel → **GPyTorch** SVGP for scalable inference. All components exist; no component needs to be built from scratch.

TDA on cortical surfaces for epilepsy is entirely unexplored

Topological data analysis in neuroimaging traces back to **Chung, Bubenik, and Kim (IPMI 2009)** — the first paper applying persistent homology to brain imaging — which used sublevel set filtration on cortical thickness data PubMed for autism analysis. Chung's group later extended this to white matter morphometry Academia.edu (IEEE TMI 2015, ~190k mesh vertices) and, most relevantly, to **functional brain networks in temporal lobe epilepsy** (NeuroImage 2023), developing Wasserstein distance-based statistical inference for persistence diagrams comparing **101 TLE patients vs. 50 controls**. PubMed Central

The broader TDA-in-neuroimaging landscape is dominated by functional connectivity applications: persistence of Rips filtrations on correlation matrices University of Florida (Petri et al. 2014, Sizemore et al. 2018), with recent work using giotto-tda on HCP data PubMed Central (Frontiers 2025). EEG seizure detection using persistent homology achieves ~99% accuracy with VR complex filtrations fed to deep networks. PubMed Central But **TDA applied directly to cortical surface meshes in epilepsy does not exist.** All epilepsy TDA work uses EEG or fMRI-derived networks, not geometric features of the cortical manifold itself.

The tooling is ready: **giotto-tda** provides scikit-learn-compatible persistence computation (JMLR) (already used in brain imaging), **GUDHI** supports sublevel set filtrations on simplicial complexes, (PubMed Central) and **torch-topological** enables differentiable persistent homology for integration with PyTorch deep learning pipelines. TopologyLayer (AISTATS 2020) provides `LevelSetLayer2D` for differentiable filtrations on surfaces. The missing piece is a dedicated sublevel-set-filtration-on-cortical-mesh pipeline — CorticalFields would need to build a lower-star filtration from FreeSurfer meshes with vertex-wise scalar functions (thickness, curvature), which is mathematically straightforward but has no existing implementation.

Bayesian MSN construction has never been attempted

Morphometric Similarity Networks (Seidlitz et al., Neuron 2018) construct brain graphs from inter-regional Pearson correlations of morphometric feature vectors, and have been actively applied to epilepsy: a 2025 study in *Acta Epileptologica* found widespread MSN alterations in mTLE linked to neuroinflammatory gene transcription. The state-of-the-art improvement is **MIND** (Sebenius et al., Nature Neuroscience 2023), which replaces Pearson correlation with KL divergence between multivariate feature distributions, improving reliability and cytoarchitectonic alignment. (Nature)

No paper constructs MSNs using Bayesian estimation of correlations. The closest methodological precedent is Janssen, Hinne, Heskes & van Gerven (Frontiers in Computational Neuroscience 2014), who obtained posterior distributions over graph metrics from structural connectivity networks estimated via Bayesian inference on diffusion tractography — but for DWI connectomes, not morphometric similarity. Replacing `np.corrcoef()` with posterior sampling from conjugate Normal-Inverse-Wishart priors would yield distributions over MSN edge weights, enabling credible intervals for every graph-theoretic metric (degree, modularity, clustering coefficient). This is computationally heavier but embarrassingly parallelizable and would provide the first uncertainty-aware structural brain network construction from T1w data.

MTLE-HS shows widespread cortical changes that no tool quantifies individually

ENIGMA-Epilepsy (Whelan et al., Brain 2018; **2,149 patients, 1,727 controls**) mapped cortical thinning in MTLE-HS with effect sizes of d = −0.29 to −0.54, affecting temporopolar, parahippocampal, entorhinal, fusiform, precuneus, precentral, and superior frontal regions. PubMed +2 Left and right MTLE show **distinct** cortical profiles: PubMed left MTLE involves more widespread temporal-limbic thinning while right MTLE shows more circumscribed patterns. DOI +2 Network-based atrophy modeling (Larivière et al., Science Advances 2020) demonstrated that atrophy preferentially affects cortical hub regions, with disease epicenters in temporo-limbic cortices. PubMed Science

Beyond thickness, subcortical findings include thalamic volume loss (d = −0.24 to −0.73, Nature preferentially mediodorsal segments), **contralateral amygdala hypertrophy** (not atrophy — a surprising finding from Neurology 2024), ResearchGate Neurology and progressive atrophy correlated with disease duration ResearchGate (meta-analytic r = −0.42 for ipsilateral hippocampus, Caciagli et al., Neurology 2017). PubMed Central PubMed Gyrification changes include increased temporolimbic cortical folding complexity PubMed Central (Bernasconi lab 2011), potentially reflecting neurodevelopmental origins.

The best automated tools are hippocampus-focused. **AID-HS** (Ripart, Wagstyl et al., Annals of Neurology 2025) achieves **90.1% sensitivity** for unilateral HS detection and **79.2% in MRI-negative cases** using HippUnfold surface features. PubMed Central **MELD** (Ripart et al., JAMA Neurology 2025) uses graph neural networks for focal cortical dysplasia detection PubMed with 34 surface features across 23 centers. Neither provides probabilistic cortical anomaly maps, and neither addresses the extrahippocampal cortical changes that ENIGMA has thoroughly documented at the group level.

The clinical need is explicit. Mito et al. (Brain 2025) reviewed normative modeling for epilepsy and called for z-score deviation maps as "the path forward" for individual-level lesion detection. Oxford Academic Current clinical practice relies on visual inspection by epileptologists, NCBI supplemented by NeuroQuant hippocampal volumetry Openaccessjournals — **20% of HS cases are missed visually**, Epilepsy Sparks +2 and extrahippocampal changes are not assessed systematically.

The PyTorch ecosystem can support the full pipeline today

The technical building blocks form a complete chain:

- **Surface I/O and processing**: nibabel (FreeSurfer/GIfTI), trimesh, PyTorch3D
- **Mesh Laplacian**: robust-laplacian (Sharp & Crane) GitHub or LaPy, both outputting sparse scipy matrices GitHub compatible with ARPACK eigensolvers PyPI
- **Spectral Matérn kernels**: GeometricKernels with PyTorch backend and GPyTorch frontend GitHub
- **Scalable GP inference**: GPyTorch's VariationalStrategy with InducingPointKernel; KeOps for large-scale kernel MVMs GitHub
- **Geometric deep learning**: DiffusionNet (Sharp et al., ACM ToG 2022) ACM Digital Library applied to cortical parcellation in Cortex-Diffusion (MICCAI 2024, handles **22k–185k vertices**); Springer PyTorch Geometric for graph convolutions on mesh data
- **Manifold optimization**: geoopt provides Riemannian SGD/Adam in PyTorch
- **TDA**: torch-topological for differentiable persistent homology; giotto-tda for classical persistence computation
- **Bayesian inference**: NumPyro or PyMC for posterior sampling of correlation matrices in Bayesian MSN

No existing library integrates these into a neuroimaging-specific pipeline. The closest is **BrainStat** (a follow-up to BrainSpace for cortical statistics) Nature +2 and the **PCNtoolkit** for normative modeling, but neither supports GP inference with manifold kernels.

Positioning CorticalFields: five layers of novelty

The gap analysis across all ten areas reveals that CorticalFields' novelty is not a single contribution but a **stack of five independently novel components**, each occupying genuine white space:

1. **GP normative modeling with geodesic Matérn kernel on cortical surfaces** — the spectral kernel theory exists (Borovitskiy 2020), arXiv normative modeling exists (PCNtoolkit), and cortical surface analysis exists (FreeSurfer), but nobody has connected all three. This is the core contribution.
2. **Information-theoretic surprise maps on cortical geometry** — no method computes vertex-wise negative log-predictive density from a spatially-aware probabilistic model on the cortical manifold. Current z-scores assume independence; deep learning reconstruction errors lack probabilistic calibration.
3. **HKS/WKS as vertex-wise cortical biomarkers** — a decade of geometry processing theory has produced rich per-vertex shape descriptors that clinical neuroimaging has almost completely ignored. Bringing these to brain surfaces for anomaly detection crosses a clear disciplinary boundary.
4. **Persistent homology on cortical surface meshes for epilepsy** — TDA has been applied to brain networks PLOS and cortical thickness (once, for autism), but never to cortical surface geometry in epilepsy. Sublevel set filtration of morphometric features on the cortical mesh is the natural construction.
5. **Bayesian MSN construction with uncertainty propagation** — all existing MSNs use point estimates (Pearson correlation or KL divergence). PubMed Central Posterior distributions over edge weights and graph metrics have methodological precedent in diffusion tractography (Janssen et al. 2014) but have never been applied to morphometric similarity.

The closest competitor to CorticalFields overall is not a single tool but a combination: **SNM** (for vertex-wise normative z-scores via eigenmodes), **MELD** (for supervised cortical surface anomaly detection via GNN), and **AID-HS** (for hippocampal HS detection). CorticalFields differentiates itself through **principled uncertainty quantification**, unsupervised anomaly detection via proper probabilistic scoring, spatial covariance modeling on the cortical manifold, and integration of spectral shape descriptors and topological features that none of these tools provide.

Conclusion

CorticalFields occupies a convergence point where mature mathematical theory (spectral Matérn GPs), NeurIPS production-ready computational infrastructure (GeometricKernels + GPyTorch + LaPy), and urgent clinical need (individual-level extrahippocampal anomaly detection in MTLE-HS) all exist but have never been connected. The library's most defensible novelty is not any single technique but the integration: a GP that respects cortical geometry, produces calibrated surprise scores, incorporates spectral shape descriptors from geometry processing, and delivers vertex-wise individual-level anomaly maps for a clinical population where group-level findings are extensive but individual-level tools are absent. The primary engineering challenge is scalability at 150k vertices — solvable via spectral truncation ($L \leqslant 1000$ eigenpairs) combined with variational sparse GP inference ($m \leqslant 1000$ inducing points) ScienceDirect — rather than any fundamental theoretical barrier. Every component has been demonstrated independently; CorticalFields' contribution is the pipeline.